NYCU IEE ICLAB – 2026 Spring

Final Project

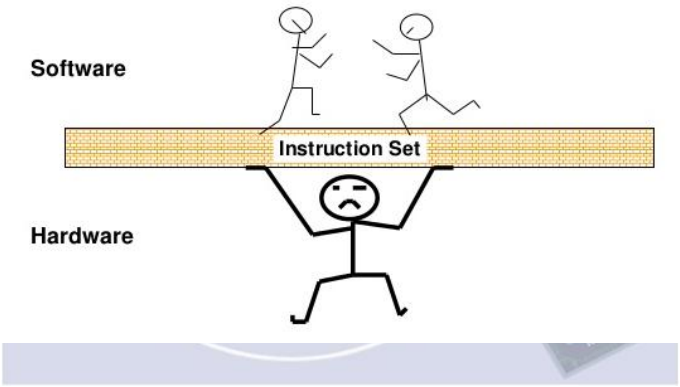
Design: Customized ISA Processor

Data Preparation

1. Extract test data from TA's directory:
   % tar xvf ~iclabTA01/FP_2026_SPRING.tar

Design Descripti

A reduced instruction set computer (RISC) is a computer instruction set (AKA: the instruction set architecture (ISA)) which allows a computer's microprocessor to have fewer cycles per instruction (CPI) than a complex instruction set computer (CISC) [from wikipedia]. In this exercise, you are asked to design a 16-bits CPU with 8 registers and the CPU should be available for the required instruction set, which shows below (very similar to MIPS).



The following are the four formats for the core instruction set:

Type	-15-	- Format – (16-bits)				-0-
R	opcode (3-bits)	rs (3-bits)	rt (3-bits)	rd (3-bits)	func (3-bits)	?
Mult	opcode (3-bits)	rs (3-bits)	rt (3-bits)	rd (3-bits)	rl (3-bits)	?
I	opcode (3-bits)	rs (3-bits)	rt (3-bits)	immediate (7-bits)		
J	opcode (3-bits)	Address (13-bits)				

Register s (rs), Register t (rt), Register d (rd) and Register l (rl) represent the address of registers. Since the instruction takes 3 bits to store the address, it means we have 8 registers, from r0 to r7. Each register reserve 16 bits to store data, e.g. rs = 3'b000 means one of operands is "r0". rt = 3'b101 means one of operands is "r5", and so on.

Detailed instructions are shown below:

Function Name	Meaning	Type	Instruction Binary Encode	Mode
ADD	rd = rs + rt	R	000-rs-rt-rd-000?	0
SUB	rd = rs – rt	R	000-rs-rt-rd-001?	1
AND	rd = rs & rt (bit-wise)	R	000-rs-rt-rd-010?	2
OR	rd = rs | rt (bit-wise)	R	000-rs-rt-rd-011?	3
NAND	rd = ~(rs & rt) (bit-wise)	R	000-rs-rt-rd-100?	4
NOR	rd = ~(rs | rt) (bit-wise)	R	000-rs-rt-rd-101?	5
XOR	rd = rs ^ rt (bit-wise)	R	000-rs-rt-rd-110?	6
Set less than	if (rs < rt) rd = 1 else rd = 0	R	000-rs-rt-rd-111?	7
Mult	{rd, rl} = rs * rt	Mult	001-rs-rt-rd-rl?	8
ADDI	rt = rs + immediate (sign)	I	010-rs-rt-iiiiiii	9
SUBI	rt = rs – immediate (sign)	I	011-rs-rt-iiiiiii	10
Load	rt = DM[sign(rs+immediate)× 2 + offset]	I	100-rs-rt-iiiiiii	11
Store	DM[sign(rs+immediate)× 2 + offset] = rt	I	101-rs-rt-iiiiiii	12
Branch on equal	If (rs == rt) pc = pc + 1 + immediate(sign) else pc = pc + 1	I	110-rs-rt-iiiiiii	13
Jump	Next instruction address = address	J	111-address	14

*?: useless bit (don't care).

*pc: program counter, meaning the next instruction. For R-type, Mult-type, and I-type instructions, pc should automatically +1 to get the next instruction.

*Each instruction has 16bits, which means the instruction address should +2 to get the next instruction.

*The offset of INST DRAM = 0x0000; the offset of DATA DRAM = 0x1000

*The first instruction address you acquire data should be "0x00000000".

```
@0000                                                    @00001000
a1 80 // I_inst: load 1000000010100001 rs = 0, rt = 1, imm =      33  65 00 // 101
@0002                                                    @00001002
3c 81 // I_inst: load 1000000100111100 rs = 0, rt = 2, imm =      60  23 85 //-31453
@0004                                                    @00001004
9d 81 // I_inst: load 1000000110011101 rs = 0, rt = 3, imm =      29  d2 00 // 210
```

Two instruction DRAM (inst_1_dram_file.dat and inst_2_dram_file.dat) and one data DRAM (data_dram_file.dat) are used in this lab. There are two cores in your design. One is used to decode instruction from instruction DRAM #1, and the other is used to decode instruction from instruction DRAM#2. Because there is only one data DRAM in this design, you need to make sure that the data you take from DRAM is newest one. You should be careful if you use the pipelining architecture, e.g. basic five-stage pipeline in a RISC machine, you must deal with data hazard problem.

Inputs & Output

I/O	Bit	Signal Name	Description
Input	1	clk	Positive edge trigger clock within clock period below 20.0ns
Input	1	rst_n	Asynchronous reset active low reset
Output	1	stall_1	Pull high when Core 1 is busy. It should be low for one cycle whenever you finished an instruction. TA will check the values of your registers of Core1 when your stall_1 is low. (refer to appendix to see how to check the register value in PATTERN)
Output	1	stall_2	Pull high when Core 2 is busy. It should be low for one cycle whenever you finished an instruction. TA will check the values of your registers of Core2 when your stall_2 is low. (refer to appendix to see how to check the register value in PATTERN)

- All inputs will be changed at clk ***negative*** edge.
- There is ***only 1 reset*** before the first pattern, thus, your design must be able to reset automatically.
- The test pattern will check the value in all registers of two cores at clock ***negative*** edge if stall is low.
- The test pattern will check the value in data DRAM every 50 instructions at clock negative edge if the last stall signal (stall_1 and stall_2) is low.
- All the registers of two cores should be zero after the reset signal is asserted.
- All the stall signals should be logic-high after the reset signal is asserted.
- ~~The value in all registers should be **unchanged** when stall signal is low.~~

AXI 4 Interface (Connected with DRAM in Pattern)

In this project, the signal naming rule for AXI4 related signals is: AXI4 signal name (lower case) + _m_inf (suffix).

Write Address Channel

Signal	Source	Description
AWID[3:0]	Master	Write address ID. This signal is the identification tag for the write address group of signals. (In this project, we only use this to recognize master, reordering method is not supported)
AWADDR[31:0]	Master	Write address. The write address bus gives the address of the first transfer in a write burst transaction. The associated control signals are used to determine the addresses of the remaining transfers in the burst.
AWLEN[6:0]	Master	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
AWSIZE[2:0]	Master	Burst size. This signal indicates the size of each transfer in the burst. (In this project, we only support 3'b001 which is 2 Bytes (matched with Data Bus-width) in each transfer)
AWBURST[1:0]	Master	Burst type. The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated. (only INCR(2'b01) is supported in this Project)
AWVALID	Master	Write address valid. This signal indicates that valid write address and control information are available: 1 = address and control information available 0 = address and control information not available. The address and control information remain stable until the address acknowledge signal, **AWREADY**, goes HIGH.
AWREADY	Slave	Write address ready. This signal indicates that the slave is ready to accept an address and associated control signals: 1 = slave ready 0 = slave not ready.

Write Data Channel

Signal	Source	Description
WDATA[15:0]	Master	Write data. The write data bus can be 8, 16, 32, 64, 128, 256, 512, or 1024 bits wide. (In this project, we only support 16 bit data width: WDATA[15:0])
WLAST	Master	Write last. This signal indicates the last transfer in a write burst.
WVALID	Master	Write valid. This signal indicates that valid write data and strobes are available: 1 = write data and strobes available 0 = write data and strobes not available.
WREADY	Slave	Write ready. This signal indicates that the slave can accept the write data: 1 = slave ready 0 = slave not ready.

Write Response Channel

Signal	Source	Description
BID[3:0]	Slave	Response ID. The identification tag of the write response. The BID value must match the AWID value of the write transaction to which the slave is responding.
BRESP[1:0]	Slave	Write response. This signal indicates the status of the write transaction. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR. (only OKAY(2'b00) is supported in this Project)
BVALID	Slave	Write response valid. This signal indicates that a valid write response is available: 1 = write response available 0 = write response not available.
BREADY	Master	Response ready. This signal indicates that the master can accept the response information. 1 = master ready 0 = master not ready.

Read Address Channel

Signal	Source	Description
ARID[3:0]	Master	Read address ID. This signal is the identification tag for the read address group of signals. (In this project, we only use this to recognize master, reordering method is not supported)
ARADDR[31:0]	Master	Read address. The read address bus gives the initial address of a read burst transaction.
ARLEN[6:0]	Master	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
ARSIZE[2:0]	Master	Burst size. This signal indicates the size of each transfer in the burst. (In this project, we only support 3'b001 which is 2 Bytes (matched with Data Bus-width) in each transfer)
ARBURST[1:0]	Master	Burst type. The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated. (only INCR(2'b01) is supported in this Project)
ARVALID	Master	Read address valid. This signal indicates, when HIGH, that the read address and control information is valid and will remain stable until the address acknowledge signal, **ARREADY**, is high. 1 = address and control information valid 0 = address and control information not valid.
ARREADY	Slave	Read address ready. This signal indicates that the slave is ready to accept an address and associated control signals: 1 = slave ready 0 = slave not ready.

Read Data Channel

Signal	Source	Description
RID[3:0]	Slave	Read ID tag. This signal is the ID tag of the read data group of signals. The RID value is generated by the slave and must match the ARID value of the read transaction to which it is responding.
RDATA[15:0]	Slave	Read data. The read data bus can be 8, 16, 32, 64, 128, 256, 512, or 1024 bits wide. (In this project, we only support 16 bit data width: RDATA[15:0])
RRESP[1:0]	Slave	Read response. This signal indicates the status of the read transfer. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR. (only OKAY(2'b00) is supported in this Project)
RLAST	Slave	Read last. This signal indicates the last transfer in a read burst.
RVALID	Slave	Read valid. This signal indicates that the required read data is available and the read transfer can complete: 1 = read data available 0 = read data not available.
RREADY	Master	Read ready. This signal indicates that the master can accept the read data and response information: 1= master ready 0 = master not ready.

Specifications

1. Top module name: **DCCPU** (top file name: **DCCPU.v**)
2. It is **asynchronous** reset and **active-low** architecture. **rst_n** would active once at the beginning. All registers of two cores should be zero after the reset signal asserts; all stall signals should be one after the reset signal asserts.
3. **All D-flip-flop (DFF) in this design must be reset after the reset signal asserts.**
4. All the data are in **signed** format in all registers.
5. The maximum clock period is set to **20 ns**, and **you can determine the clock period by yourself**.
6. The input delay and the output delay should be **half** of the clk period, if clock period is 10ns, the input delay and the output delay will be 5ns.
7. The stall signals cannot be continuous high for **1415** during **performance check**.
8. The output loading is set to **0.05**.
9. The synthesis result of data type cannot include any **LATCH** (in syn.log).
10. In this project, you should **modify the file of .synopsys_dc.setup in 02_SYN (e.g. link library for your memory.db) by yourself**. **compile_ultra** will be used to synthesize.

11. The memory.db file for synthesis should be the WC library.
12. You **CANNOT PASS** the demo if there are **timing violation** messages in the gate level simulation WITHOUT notimingchecks option
13. No **ERROR** is allowed in every simulation/synthesis.
14. **You must use SRAM in your design or you wouldn't get any point in this project.**
15. The maximum area of 02_SYN should be smaller than **800,000.**
16. **You cannot change the register name or type from the original file.**
17. PATTERN will not generate any overflow cases during calculation.
18. For achieving the data dependence in real case, the next load/store address will be constrained in certain range for data memory and the address for jump/branch instruction will also be constrained in certain range.
19. The latency between BVALID and BREADY, RREADY and RVALID should not be larger than **300** cycles.
20. DRAM must be declared in PATTERN, **do not declare it in your DESIGN.**
21. Don't use any wire/reg/submodule/parameter name called *error*, *latch*, *congratulation* or *fail* otherwise you will fail the lab. Note: * means any char in front of or behind the word. e.g: error_note is forbidden.
22. Don't write Chinese comments or other language comments in the file you turned in.
23. Designware IPs are forbidden in this project.
24. Any form of display or printing information in verilog design is forbidden. You may use this methodology during debugging, but the file you turn in should not contain any coding that is not synthesizable.

APR Specifications

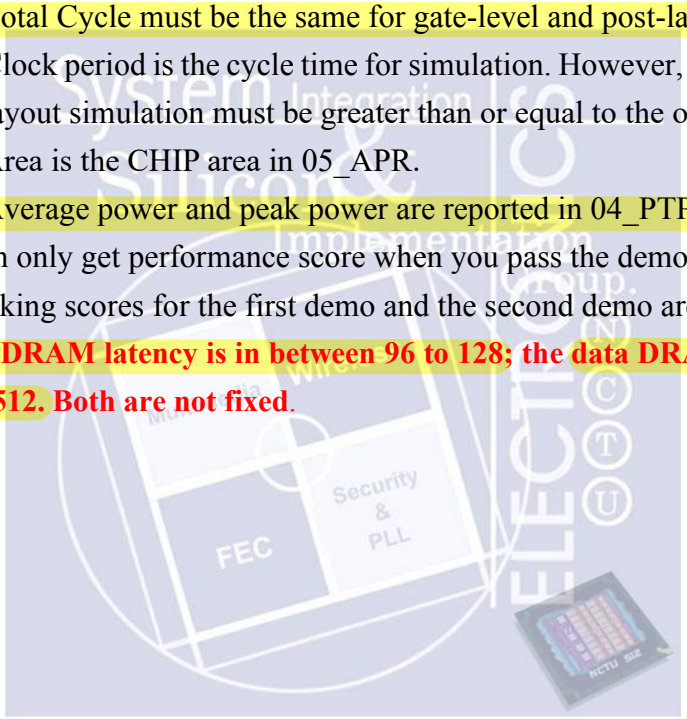
1. Floorplan:
   a. The core utilization must be larger than 0.5.
   b. Core to IO boundary must be meet the minimum space for power ring.
   c. Hard macro must be placed inside core.
2. Powerplan:
   a. The width of each individual metal wire in the power ring must be greater than or equal to 9, and strictly less than 10.
   b. The power ring must utilize wire group and interleaving, and the number of groups must satisfy the peak power requirements.
      In the Final Project, assuming that a metal wire can withstand 1 mA of current per 1 μm of width in the U18 process:
      Given the current wire width is 9 μm and the number of bits per wire group is 10, this power ring configuration can supply a maximum power of:
      $$9\ \text{mA} \times 10 \times 1.8\ \text{V} = 162\ \text{mW}$$
      This calculated value must strictly exceed the peak power reported in 04_PTPX.
   c. Stripes sets distance should be less than 200, width at least 4.
3. Timing slack shouldn't have any negative slacks after setup/hold time analysis (including SI).
4. The DRV of (fanout, cap, tran) should be all 0 after post_Route setup/hold time analysis (including SI).
5. No DRC violation after "verify geometry".
6. No connectivity violation after "verify connectivity".
7. Core filler must be added.

Grading Policy

- RTL, Synthesis, Gate-Level & Post Simulation Correctness: 50%
  You can only get this score and pass the demo with correct RTL, synthesis, gate-level and post simulation.
  There is no hidden case in this Final Project. However, if you fail before the stage of APR, you would lose the chance of 2nd Demo.
- Performance: 50%
  1. **Performance:**
     - **Total Cycle x Clock Period x Area: 25%**
     - **Total Cycle x Clock Period x Average Power x Peak Power: 25%**

     Note:
     - Total Cycle must be the same for gate-level and post-layout simulation.
     - Clock period is the cycle time for simulation. However, the one for post-layout simulation must be greater than or equal to the one for synthesis.
     - Area is the CHIP area in 05_APR.
     - Average power and peak power are reported in 04_PTPX. (Unit: W)
  2. You can only get performance score when you pass the demo.
  3. The ranking scores for the first demo and the second demo are separate.

The instruction DRAM latency is in between 96 to 128; the data DRAM latency is in between 256 to 512. Both are not fixed.



mem.db 都放在 4 個 .db 之後

04 PT 也要放 mem.db

Note

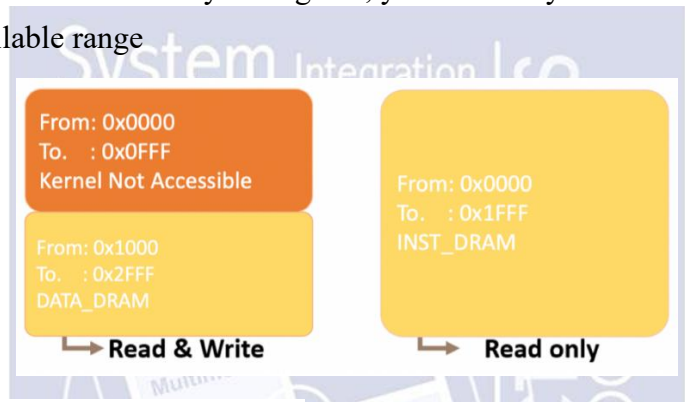
1. Please submit your design file under the folder 09_SUBMIT before **12:00 noon**.
   Due Day : 1st Demo : June 24th 2nd Demo : June 26th
   In this lab, you can adjust your cycle time.
   After that, you should check the following files under
   09_SUBMIT/Final_Project_iclabXXX/
   RTL design: **DCCPU**_iclabXXX.v (XXX is your account no.)
   GATE_clock_cycle_iclabXXX.txt
   POST_clock_cycle_iclabXXX.txt
   Memory file: 04_MEM_iclabXXX
   (with all your memory files .v, .db, .lef & .lib file)
   filelist_iclabXXX.f
   APR and Post design:
   - CHIP_iclabXXX.sdc
   - CHIP_cts_iclabXXX.sdc
   - CHIP_iclabXXX.inn
   - CHIP_iclabXXX.io
   - CHIP_iclabXXX.v
   - CHIP_iclabXXX.sdf
   - CHIP_iclabXXX.inn.dat
   - mmmc_iclabXXX.view

   If you miss any files on the list, you will fail this lab.
   If any error occurs when restoring your design, you will FAIL the project.
   Then use the command like ./02_check to check the files are uploaded or not.
2. Template folders and reference commands:
   01_RTL/ (for RTL simulation) **./01_run_vcs_rtl**
   02_SYN/ (for Synthesis) **./01_run_dc_shell**
   Remember modify .synopsys_dc.setup to fit your memory .db name
   (Check the design which contains **Latch** and **Error** or not in **syn.log**)
   (Check the design's timing in /Report/***DCCPU.timing*** to see if the slack is **MET**)
   03_GATE / (Gate Level simulation) **./01_run_vcs_gste**
   You can key in **./09_clean_up** to clear all log files and dump files in each folder.
   04_MEM/ (Memory location)
   You should generate your memories and put the required files (.v and .db) here.
   04_PTPX/ (PrimeTime) **./01_run_ptpx**
   You should run power analysis here.
   05_APR/ (APR folder)
   You should do the APR for you design here.
   06_POST / (Post-layout simulation) **./01_run_vcs_post**
   You can key in **./09_clean_up** to clear all log files and dump files in each folder
   09_SUBMIT/ (Submit your files)
   ./00_tar ./01_submit ./02_check

Appendix

- TA will check the value in data memory **every 50 instructions and at the end of the simulation** when the last stall signal is low.
    - The Instruction Counter (IC) starts counting from 0. Therefore, the 50th instruction refer to the instruction executed when the IC is 49.
    - The simulation will terminate when two conditions meet:
        - The instruction at the last address is reached.
        - Both cores have executed the exact same number of instructions.
- TA will check your registers of cores in PATTERN when the stall signals of the cores are low.

  Please make sure your registers can be read by PATTERN after synthesis, otherwise if TA's pattern cannot access your register, you will fail your demo.
- Memory available range



- Store, load, branch and jump address

  The address value of the above instructions must be within the available address range for DRAM.
- The sequence of instructions between the two cores is strictly determined by the Instruction Counter (IC) for this project. For example, the 19th instruction of Core 1 is guaranteed to be executed before the 20th instruction of Core 2. Please pay close attention to the execution efficiency of both cores under this constraint.
- When both cores are at the same Instruction Counter (IC) value, the following rules apply to Load/Store operations:
    - **Store-before-Load Priority**: If Core 1 performs a Load instruction and Core 2 performs a Store instruction at the same IC, the Store operation must be completed first. For example, at IC = 20, if Core 2 stores to an address that Core 1 is loading from, Core 1 will receive the updated value from Core 2's store.
    - **Exclusive Store Access**: At any given IC, if one core performs a Store operation, the other core is strictly prohibited from performing any Store operation, regardless of the target address.

- Instruction

  Ex. 000-0001-0010-0011-0 means r1 + r2 and saves the result into r3.

  The first instruction address you acquire data should be "0x00000000".

- Data Dependence Prediction

  For achieving the data dependence in real case, the next load/store address of cores will be constrained in certain range for one data DRAM and the address for jump/branch instruction will also be constrained in certain range. The range of data address is from (current address – 200) to (current address + 200). The range of jump instruction is from (current address – 256) to (current address + 256), e.g. if current pc is dec(1000), the next instruction pc will be from dec(872) to dec(1128).

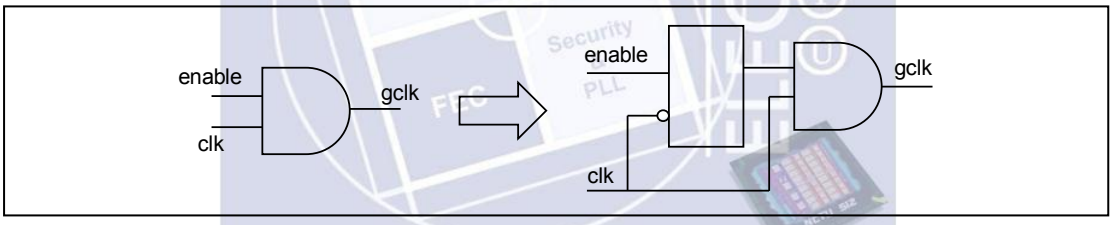
- Manual Clock Gating

  Coding style in RTL:

```
assign gclk = clk & enable;
 enable =1 的時候才會改值
always @(posedge gclk or negedge rst_n) begin
      if (!rst_n) begin
            D <= 'd0;
      end else begin
            D <= D + 'd1;
      end
```

  syn.tcl

```
replace_clock gates
compile_ultra
```



A careful person will eventually have good luck.